32 mm (1.26 in) bore × 16 mm (0.63 in) rod Cylinder Force Data

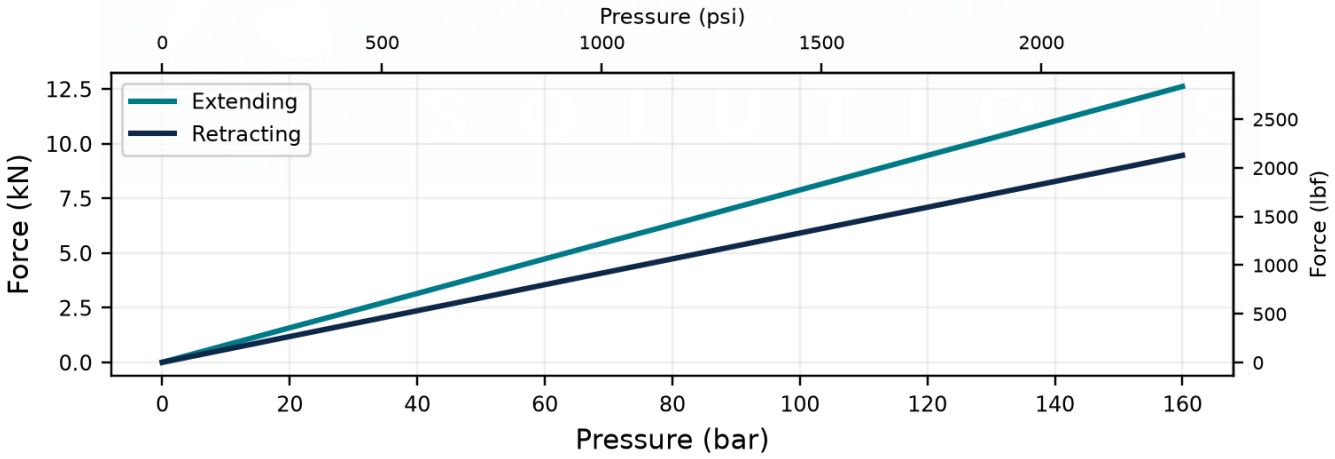
This geometry has a retraction area equal to 75.0% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 8.04 cm² (1.246 in²)	Ring Area 6.03 cm² (0.935 in²)
Max Extending Force 12.61 kN (2,835 lbf)	Max Retracting Force 9.46 kN (2,127 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	0.79 kN (178 lbf)	0.59 kN (133 lbf)
20 bar (290 psi)	1.58 kN (355 lbf)	1.18 kN (265 lbf)
30 bar (435 psi)	2.36 kN (531 lbf)	1.77 kN (398 lbf)
40 bar (580 psi)	3.15 kN (708 lbf)	2.36 kN (531 lbf)
50 bar (725 psi)	3.94 kN (886 lbf)	2.95 kN (663 lbf)
60 bar (870 psi)	4.73 kN (1,063 lbf)	3.55 kN (798 lbf)
70 bar (1,015 psi)	5.52 kN (1,241 lbf)	4.14 kN (931 lbf)
80 bar (1,160 psi)	6.30 kN (1,416 lbf)	4.73 kN (1,063 lbf)
90 bar (1,305 psi)	7.09 kN (1,594 lbf)	5.32 kN (1,196 lbf)
100 bar (1,450 psi)	7.88 kN (1,771 lbf)	5.91 kN (1,329 lbf)
110 bar (1,595 psi)	8.67 kN (1,949 lbf)	6.50 kN (1,461 lbf)
120 bar (1,740 psi)	9.46 kN (2,127 lbf)	7.09 kN (1,594 lbf)
130 bar (1,885 psi)	10.24 kN (2,302 lbf)	7.68 kN (1,727 lbf)
140 bar (2,031 psi)	11.03 kN (2,480 lbf)	8.27 kN (1,859 lbf)
150 bar (2,176 psi)	11.82 kN (2,657 lbf)	8.86 kN (1,992 lbf)
160 bar (2,321 psi)	12.61 kN (2,835 lbf)	9.46 kN (2,127 lbf)